

Network Testing Practical Report

Task 1: Testing Connectivity Using Ping

Procedure

1. Connected the laptop and a mobile phone to the same Wi-Fi network.
2. Found the IP address of the phone.
3. Opened **Command Prompt** on the laptop.
4. Executed the command:

```
ping <phone_IP_address>
```

Example:

```
ping 192.168.1.20
```

Observation

The ping command returned replies from the destination device, confirming successful communication over the local network.

Task 2: Running Traceroute

Command

```
tracert www.youtube.com
```

Observation

Traceroute displayed the path taken by packets from the local computer to YouTube, listing each intermediate router (hop).

Number of Hops

Example:

12 hops

Task 3: Ping Test After Disconnecting One Device

Procedure

1. Disconnected the mobile phone from the Wi-Fi network.
2. Ran the ping command again from the laptop.

Result

The laptop could no longer reach the mobile device.

Example Error Message

```
Request timed out.
```

or

```
Destination host unreachable.
```

Explanation

Since the phone was disconnected from the Wi-Fi network, it no longer had an active network connection, so the laptop could not communicate with it.

Task 4: Devices Connected to the Wi-Fi Network

Command Used

```
arp -a
```

Example Table

IP Address	MAC Address	Device
192.168.1.1	XX-XX-XX-XX-XX-XX	Wi-Fi Router
192.168.1.10	XX-XX-XX-XX-XX-XX	ASUS TUF A15 Laptop
192.168.1.20	XX-XX-XX-XX-XX-XX	Friend's Smartphone
192.168.1.30	XX-XX-XX-XX-XX-XX	Smart TV

Task 5: One-Page Report

Title

Setting Up and Testing a Basic Network

Report

The practical activity began by connecting an ASUS TUF A15 laptop and a mobile phone to the same Wi-Fi network. After confirming that both devices had obtained IP addresses from the router, the IP address of the mobile phone was identified.

A **ping** test was performed from the laptop to the phone using the Command Prompt. Successful replies confirmed that both devices were communicating correctly over the local area network.

Next, the **tracert** command was used to trace the route from the laptop to www.youtube.com. The command displayed each router (hop) through which the data traveled before reaching the destination server. This demonstrated how network packets move across multiple networks to reach an internet service.

To test network troubleshooting, the mobile phone was disconnected from the Wi-Fi network. When the ping command was executed again, the laptop displayed a "**Request timed out**" (or similar) message, indicating that the destination device was no longer reachable.

Finally, the **arp -a** command was executed to display the Address Resolution Protocol (ARP) table. This table listed devices currently known on the local network, along with their IP addresses and MAC addresses. The devices were identified based on their IP addresses and ownership.

Conclusion

The practical demonstrated how to establish a simple network, verify connectivity using the **ping** command, examine routing paths using **tracert**, identify connected devices using **arp -a**, and recognize common connectivity issues when a device disconnects from the network.